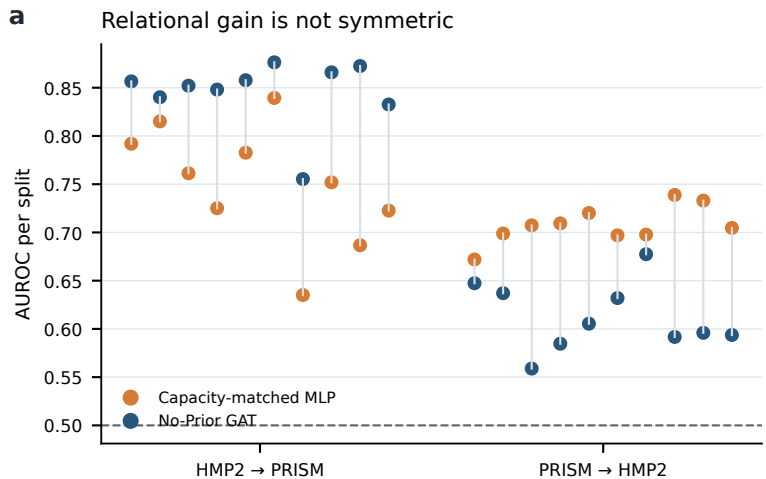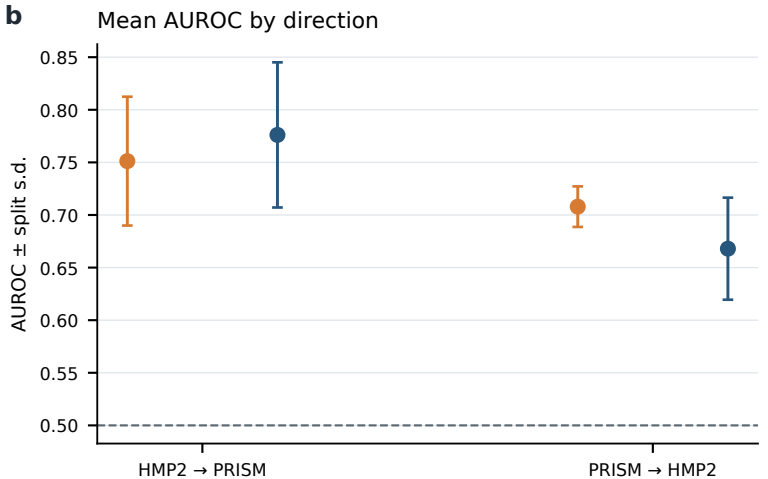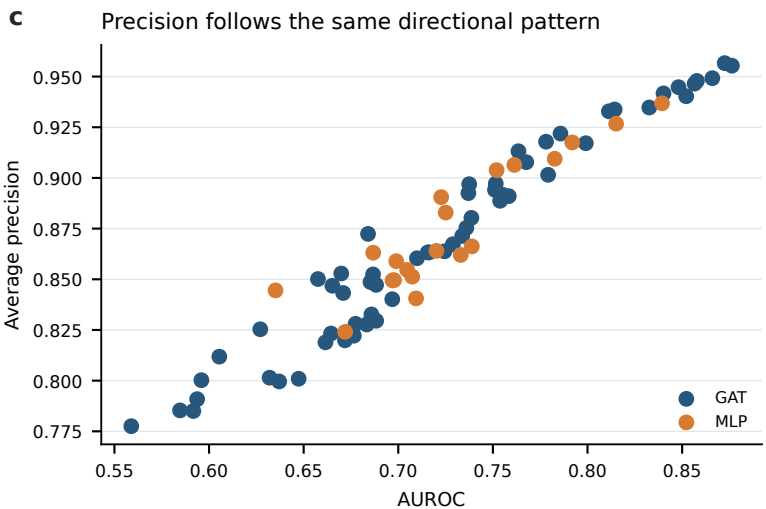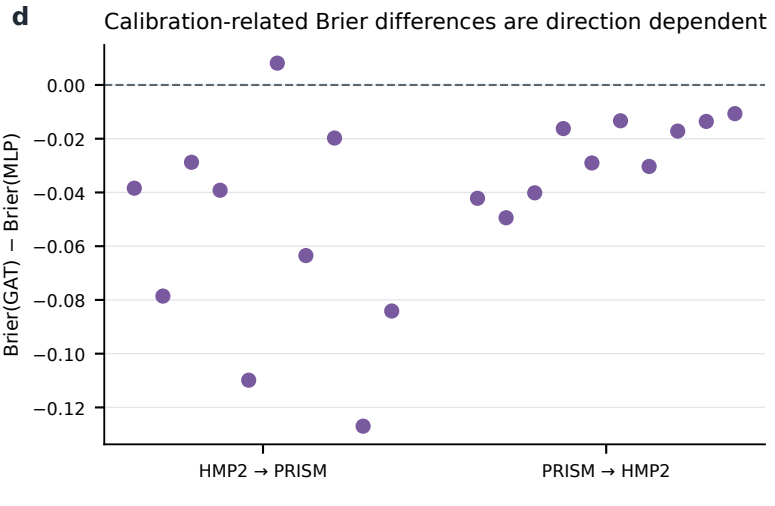Capacity-matched control separates graph message passing from generic nonlinear capacity